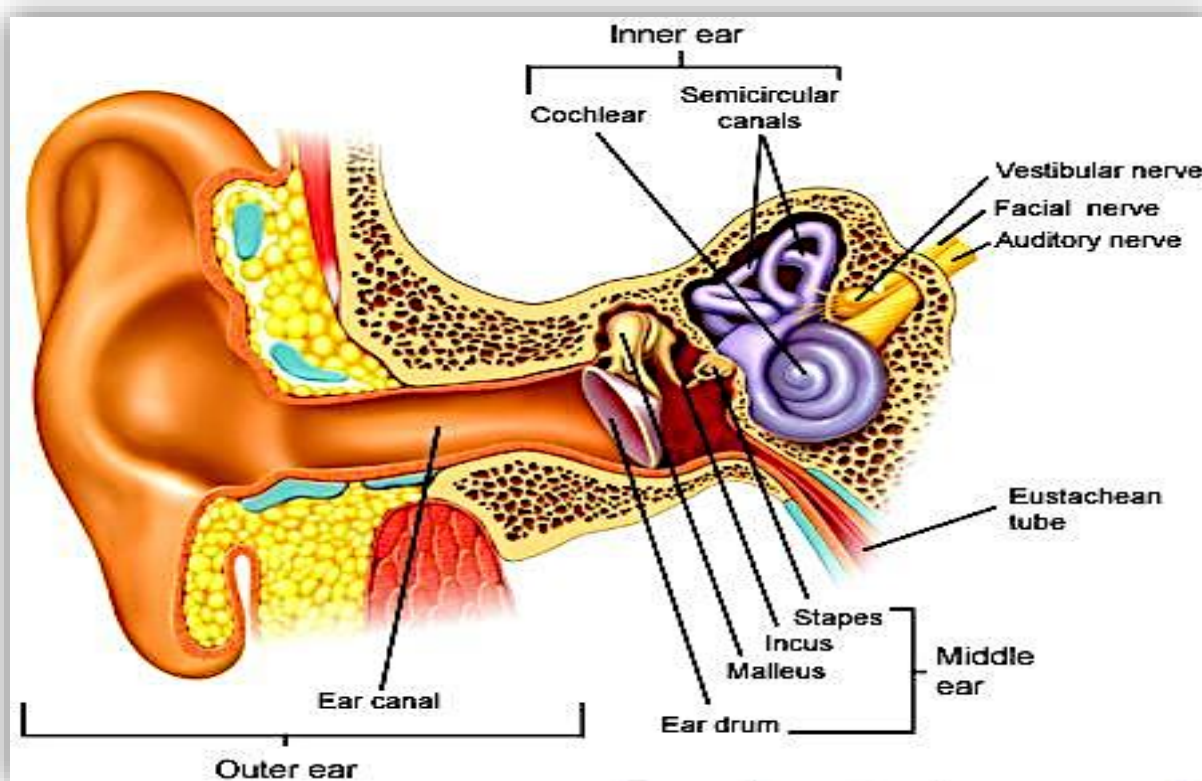


The Ear



The ear is formed of three parts; external, middle and internal ear.

1. External ear: composed of:

1. **Auricle:** elastic cartilage covered by skin
2. **External auditory meatus:** Rigid hollow tube starts from auricle to tympanic membrane. Outer third elastic cartilage, inner two thirds bone. Lined with hairy skin containing sebaceous and ceruminous glands "modified sweat glands that open in surface of meatus, they produce cerumin.
3. **Tympanic membrane "ear drum":** Dense connective tissue with collagen (mechanical strength) and elastic fibers, covered externally by thin skin and internally by low cuboidal epithelium. Function: It transmits sound vibration to the ossicles of the middle ear.

2. Middle ear "tympanic cavity":

lies in a cavity in the temporal bone between the tympanic membrane and the bony wall of the inner ear. It is communicated anteriorly with the nasopharynx by the Eustachian tube while posteriorly, it is continuous with the mastoid air sinus. It is lined by low cubical epithelium that becomes columnar ciliated at opening of auditory tube. Epithelium rest on loose connective tissue. The middle ear conducts vibration from tympanic membrane to inner ear through the bony ossicles.

Structure: 3 bony ossicles, 2 muscles and 2 windows.

- a. Ossicles: They are malleus, incus, and stapes. Formed of bone and its periosteum covered by the same epithelial lining of middle ear. They connect tympanic membrane to the oval window. They are joined to each other by regular joints.
- b. Muscles: Two striated muscles attached to ossicles:
 - Tensor tympani, attached to malleus.
 - Stapedius: attached to stapes.It plays a role in sound conduction, protection of middle ear and makes the drum tense. It also diminishes tension on middle ear.
- c. Windows:
 - Oval window: closed by foot plate of stapes. It transmits the vibration to the inner ear.
 - Round window: closed by elastic membrane "secondary tympanic membrane".

The middle ear is connected with the nasopharynx by the Eustachian (auditory) tube. This tube is lined by simple columnar ciliated epithelium near tympanic cavity that becomes pseudostratified columnar ciliated with goblet cells near nasopharynx. Its anterior 2/3 are supported by fibrocartilage while its posterior 1/3 is bony. The function of the middle ear is to equalize air pressure in tympanic cavity and nasopharynx.

Inner ear: Composed of:

- Bony labyrinth
- Membranous labyrinth
- Sensory receptor structures.

Bony labyrinth: Cavity in temporal bone that contains membranous labyrinth. Areas between bones and fluid spaces filled with fibro elastic CT. Fluid spaces lined with simple sq epithelium. Components are vestibule, semicircular canals, and cochlea.

- Vestibule: irregular central cavity contains utricle and saccules.
- Three semicircular canals: lying posteriorly and enclose semicircular ducts and open into utricle and each canal has a dilatation called ampulla.
- Cochlea: present anterolaterally contains cochlear ducts.



Membranous Labyrinth (Utricle and saccule):

Composed of CT sheath lined with simple sq epith and connected with bony labyrinth by thin strands of CT containing blood vessels. Utricle larger than saccule and both connected by utriclo-saccular ducts. Walls contain sensory receptor cells "Macula of utricle and macula of saccule".

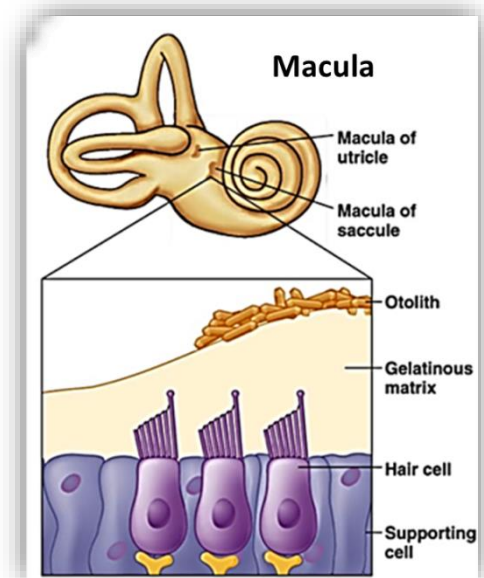
1. Macula: Thickened area of epithelium resting on basal lamina, and have two types of cells:

a. Hair cells which are of 2 types:

- Type 1: flask-shaped
- Type 2: pyramidal in shape.

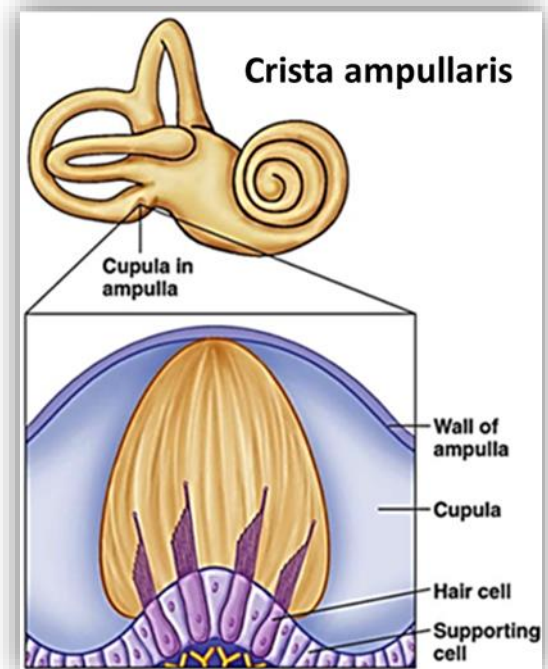
Each hair cell has free surface with elongated microvilli (stereocilia) and one non-motile cilium (kinocilium). Stereocilia are arranged in rows of increasing length with the longest located near kinocilium. Both microvilli and kinocilium embedded in a gelatinous membrane "Otolithic membrane" that contains calcium carbonate crystals called "**Otoliths**". Basal parts surrounded with afferent and efferent fibers, which are dendrites of bipolar nerve cells present in vestibular ganglia.

b. Supporting cells: Columnar cells with basal nuclei and located between hair cells. Have apical microvilli. The function of maculae is to maintain the body equilibrium in relation to the linear acceleration of the head.



2.Semicircular ducts: Have same general form as bony labyrinth. The three ducts are arranged at right angles to one another. They are continuous at both ends with utricle. Each canal contains a dilatation called ampulla in which there is a receptor area called Crista ampullaris "similar structure as macula of utricle and saccule except that the covering gelatinous membrane has no calcium carbonate crystals and is called Copula". Basal portions of hair cells are also surrounded by nerve endings, which are dendrites of bipolar nerve cells present in vestibular ganglia. Axons of these cells form vestibular division of 8th cranial nerve.

Function of crista ampullaris: It controls the body equilibrium in relation to the angular acceleration of the head which causes displacement of copula and displacement of hair cells. Impulses transmitted to nerve endings of vestibular division of 8th cranial nerve.



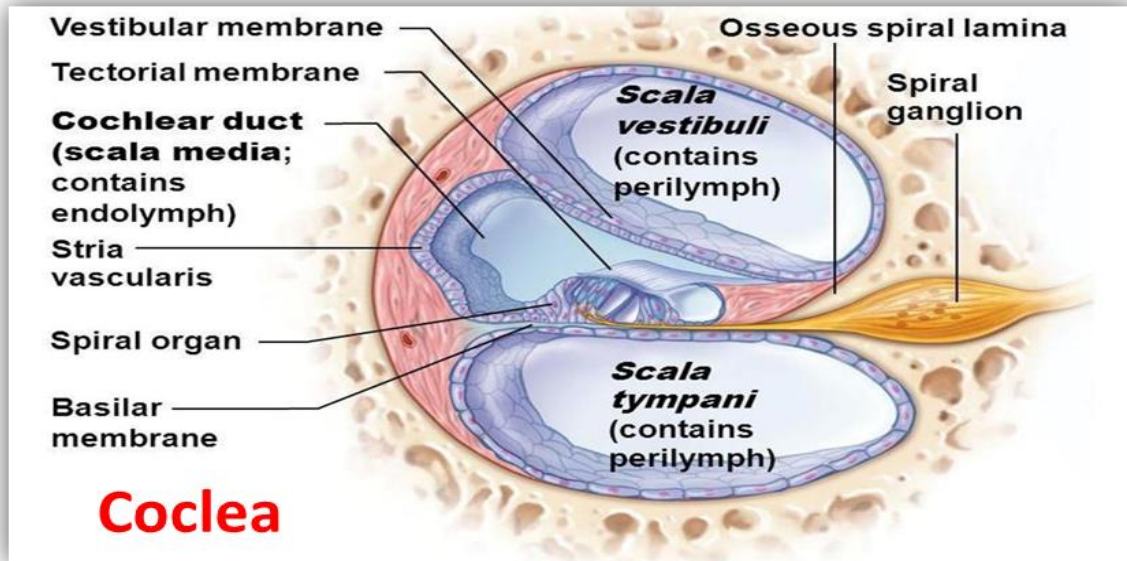
Fluid inside membranous labyrinth is called endolymph and in bony labyrinth called perilymph communicates with CSF space by narrow canal called "Vestibular aqueduct".

Cochlea:

A snail shell like tubular structure formed of that opens into anterior part of vestibule and is formed of spongy bone. **It contains a** central pillar "modiolus" around which hollow bony tube makes 2 ½ turns forming five circles. Each circle divided by 2 membranes (Vestibular membrane above and basilar membrane below) into 3 compartments:

- Scala vestibuli above,
- Scala media (cochlear duct) in middle
- Scala tympani below.

Scalae vestibuli and tympani contain perilymph and connected at tip of cochlea by an opening called "Helicotrema". Scala vestibuli communicates with vestibule and scala tympani communicates with middle ear through round window. Cochlear duct is triangular with its apex directed towards modulus, contains endolymph and ends blindly.



Walls of the cochlear duct:

- **Vestibular (Reissner's) membrane:** separates the scala vestibuli above from cochlear duct in the middle. It is formed of two layers of flattened epithelial cells one derived from cochlear duct and other from lining of scala vestibuli. The two layers are separated by a basal lamina.
- **Basilar membrane:** separates the cochlear duct from scala tympani. It is thicker than vestibular membrane and is formed of fibrous CT. The organ of Corti lies at its upper surface.
- **Stria vascularis:** it consists of a vascularized stratified columnar epithelium, resting on CT. It is present at the outer wall of cochlear duct and secretes endolymph.

Organ of Corti: It is situated in cochlear duct resting on the basilar membrane. Like macula and crista ampullaris, it is formed simply of hair and supporting cells arranged around triangular space called Tunnel of Corti.

1. Hair cells:

- **Inner:** Single row of smaller rounded cells
- **Outer:** arranged in 3-5 parallel rows of tall and slender cells.

Hair cells have stereocilia, embedded in rigid gelatinous sheet "Tectorial membrane". Their bases lie on concavities of phalangeal cells. Afferent nerve endings make synaptic contact with hair cells, which are dendrites of bipolar nerve cells present in spiral ganglion located in modiolus. Their axons form cochlear division of 8th cranial nerve.

2. Supporting cells:

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- a. Inner and outer pillar cells: Tall columnar cells with deeply stained cytoplasm. Have numerous microfilaments. Inner cells are taller than the outer cells.
- b. Inner and outer phalangeal cells: Have apical concavities to support bases of hair cells. Have supportive function.
- c. Hensen cells: Tall cells, present adjacent to last row of outer phalangeal cells.

Mechanism of hearing: Sound waves causes vibration of tympanic membrane which in turn leads to vibration of bony ossicles. These vibrations will be transmitted to the perilymph in inner ear leading to vibration of basilar membrane that causing displacement of hair cells against the tectorial membrane causing depolarization of hair cell and the impulse is transmitted by nerve fibers to the brain.

